

Unit 2, Lesson 2: Finding the Least Common Multiple

Benchmark

MA.6.NSO.3.1 Given a mathematical or real-world context, find the greatest common factor and least common multiple of two whole numbers.

Learning Target

- ☐ I can find the least common multiple (LCM) in mathematical and real-world contexts.

2.2.1 Warm-Up: Hourglass Timing

The video shows three different hourglass sand timers. The timers are all started at the same time. As one of the times runs out, it is flipped over again to restart.



Several screenshots taken from the video are shown.

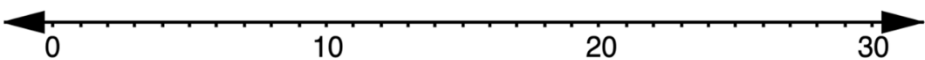
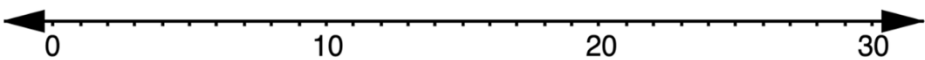
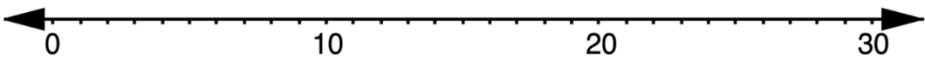


- 1. Write one question you have as you watch the video or review the images.
- 2. Compare questions with your partner.
- 3. Record your partner’s question in the table and have them initial that you recorded it correctly. If you had the same question, work with your partner to generate a different question to ask about the video. *You are the only person who should write in the first column of the table.*

My Partner’s Question From the Video	Initials

2.2.2 Guided Instruction: Least Common Multiple

1. Three timers were used in the video, a 2-, 3-, and 5-minute timer.
- a. On each number line model, plot a point indicating the time at which each hourglass would need to be flipped over to restart the timer during a 30 minute period.

Time	Points Where the Timer Needs to be Restarted
2-minute timer	
3-minute timer	
5-minute timer	

- b. What do you notice about the times plotted on the number line for each timer?

Multiples of positive whole numbers are the product of that number multiplied by a positive whole number.

For example, the **multiples** of 2 are 2, 4, 6, 8,



The **ellipses** (...) at the end of the list of multiples means that the list continues beyond the last number listed. There are an **infinite** number of multiples for any given whole number.

2. Use the number line models for each of the timers and the definition of multiples to answer the questions.

- a. What are all the common multiples among the three timers?
- b. What do these common multiples represent in this scenario?
- c. After how many minutes will the 2- and 5-minute timers run out of sand at the same time?
- d. After how many minutes will the 3- and 5-minute timers run out of sand at the same time?



The ***Least Common Multiple***, or ***LCM***, is the lowest number that is a multiple of two or more given numbers.

3. Use the definition to complete the statement.

The least common multiple (LCM) of 2, 3, and 5 is _____.

In the context of the hourglass timers, this value

represents _____

_____.

4. Find the LCM of 8 and 6.

2.2.3 Try It: Least Common Multiple

1. Find the LCM of each pair of numbers.
 - a. 10 and 4
 - b. 5 and 13
 - c. 7 and 21

2. Every 4 weeks, Fort Caroline Middle has an early release day. Every 5 weeks, they host an assembly with a guest speaker for all students. After how many weeks will Fort Caroline Middle have an assembly and early release in the same week?

3. A radio station is running two ongoing promotions. For the first promotion, every 30th caller wins a pair of movie tickets. For the second promotion, every 100th caller wins an annual pass to a theme park. Which caller number would be eligible for both prizes?

4. Vincenzo and Marie both joined a local gym. Vincenzo plans to go to the gym every 2 days at 7:00 am when they open. Marie plans to go every 3 days at 7:00 am. If they both joined the gym on the same day, after how many days will they arrive at the gym on the same day?

2.2.4 Collaboration: Least Common Multiple

1. Place a digit in each blue box to make a true statement. Use only digits 0 to 9, at most one time each.

Factor A:

Factor B:

LCM:

2. What were some of the challenges you faced with this problem?
3. What methods did you use to solve this problem?

2.2.5 Your Turn!

1. Find the LCM of each pair of numbers.
 - a. 15 and 20
 - b. 18 and 24
 - c. 22 and 11

2. Jasmine and Teia are running a race. Jasmine runs at a rate of 4 minutes per lap and Teia runs at a rate of 6 minutes per lap. If they both start running at the same time, after how many minutes will they both be at the starting line at the same time again?

3. Lin's uncle is opening a bakery. On the bakery's grand opening day, he plans to give away prizes to the first 50 customers that enter the shop. Every 9th customer will get a free blueberry muffin. Every 12th customer will get a free slice of carrot cake. Which customer will get both free bakery items?

2.2.6 Wrap-Up: Reflection

1. Rate your level of mathematical confidence on the concepts listed by circling the statement that best describes your understanding.

• **Greatest Common Factor (GCF)**

I don't know what GCF is.	I need a little more practice.	I am okay at this.	I'm pretty good at this.	I could teach this.
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• **Least Common Multiple (LCM)**

I don't know what GCF is.	I need a little more practice.	I am okay at this.	I'm pretty good at this.	I could teach this.
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